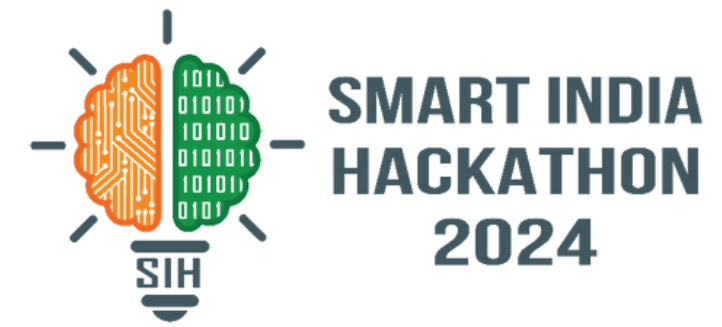


SMART *INDIA* HACKTHON



- **Problem Statement ID – SIH1572**

- **Problem Statement Title–**

Design/Development of an efficient Energy Storage System (ESS) to integrate intermittent Renewable Energy sources and to support/stabilize the grid.

- **Theme– Renewable / Sustainable Energy**

- **PS Category– Hardware**

- **Team ID–**

- **Team Name – THE RESISTORS**

AIRSHIP ENERGY STORAGE SYSTEM

RENEWABLE ENERGY STORAGE USING AIRSHIP

Detailed explanation of the proposed solution

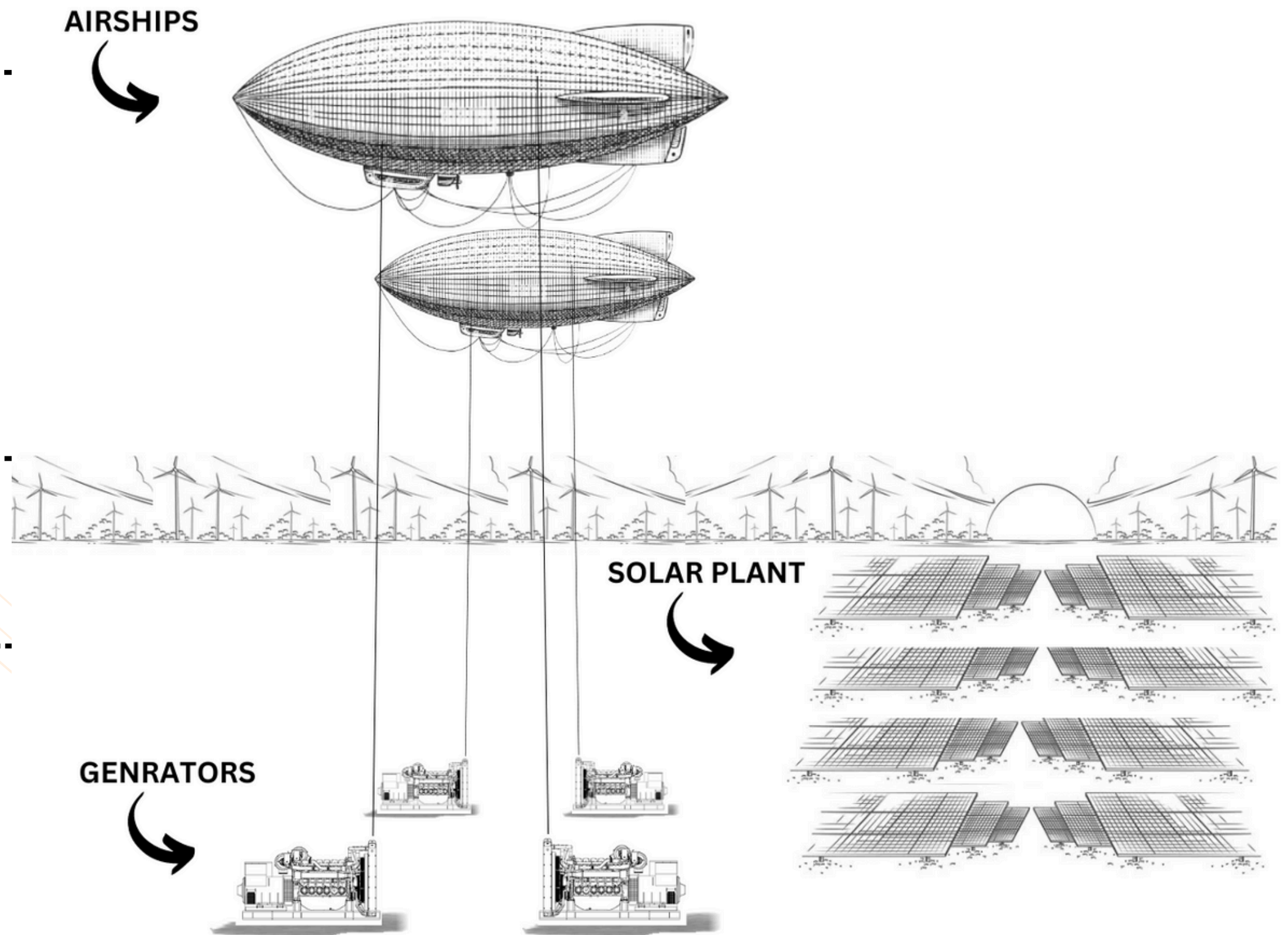
OUR AIRSHIP BASED ENERGY STORAGE SYSTEM STORE THE ELECTRICAL ENERGY IN THE FORM OF MACHANICAL POTENTIAL ENERGY BY PULL THE AIRSHIP FILLED BY THE HELIUM AGAINST THE GRAVITY BY THE USE OF EXCESS ELECTICITY PRODUCED BY THE RENEWABLE SOURCES SUCH AS SOLAR POWER PLANT AND AT THE PEAK DEMAND TIME THIS SYSTEM PROVIDE ELECTRICITY BY RELEASING AIRSHIP (CONNECTED WITH GENRATORS USING ROPES) IN THE AIR FREELY

How it addresses the problem

STORE THE ENERGY PRODUCED BY THE RENEWABLE RESOURCES AND PROVIDE THE STABILITY IN THE GRID BY ADDING ROTATION INERTIA

Innovation and uniqueness of the solution

THE UNIQUENESS OF OUR IDEA IS THAT IT IS VERY SIMPLE TO MADE AND ECONOMICALLY STABLE AS COMPARED WITH OTHER GRID SCALE ENERGY STORAGE SYSTEM (IN LONG RUN) AND THE INNOVATION IS OUR TECHNIQUE - THIS TYPE OF ENERGY STORAGE SYSTEM IS NEVER IMPLEMENTED IN THE HISTORY HERE WE ARE USING HELIUM AIRSHIP TO STORE THE ELECTRICITY AN BY THE COMBINATION OF AIRSHIPS, SIZES, PRESSURE ETC WE CAN EASILY MAINTAIN THE STORAGE OF THE ELECTRICITY

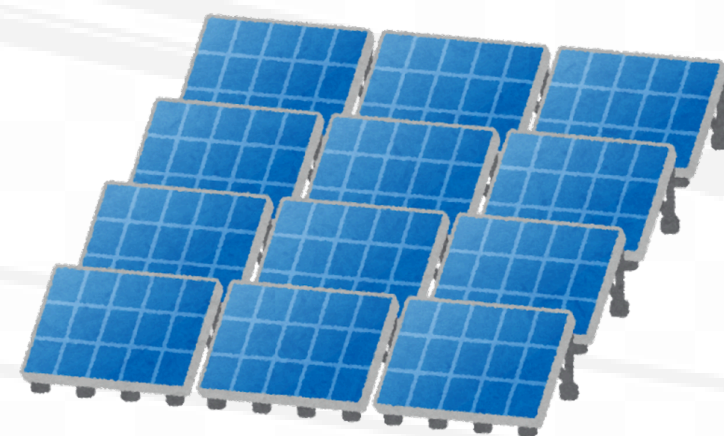


Technologies to be used (e.g. programming languages, frameworks, hardware)

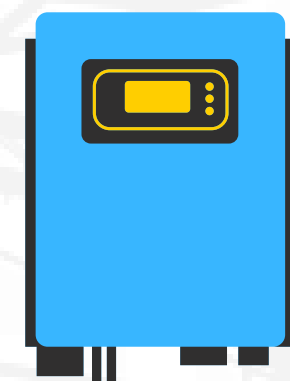
THERE IS NOTHING SPECIAL TECHNOLOGY IN THIS TYPE OF ENERGY STORAGE SYSTEM HERE WE ARE USING THE AIRSHIP TECHNOLOGY WITH IS MORE THEN 100 YEARS OLD AND WELL TESTED WE HAVE SUFFICITENT DATA ON ARISHIPS WE CAN MAEK THE AIRSHIP BASED ENERGY STORAGE SYSTEM WITH MINNOR CHANGES

Methodology and process for implementation (Flow Charts/Images/ working prototype)

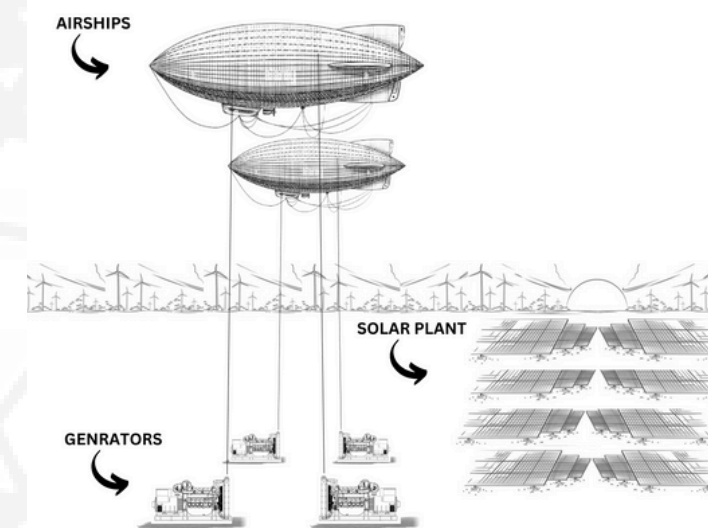
THE IMPLEMENTATION OF AIRSHIP BASED ENERGY STORAGE SYSTEM IS VERY SIMPLE WE ONLY HAVE TO MAKE THE AIRSHIP AND CONNECT THEM WITH GENERATORS IN A CONTROL MANNER NEAR ANY RENEWABLE POWER PLANT SUCH AS SOLAR POWER PLANT MAY BE THE WE CAN FACE SEVERAL CHALLENGES IN THE CASE OF CONTROLLING THE AIRSHIP IN AIR BUT IT IS POSSIBLE WITH MODERN DAY TECHNOLOGY



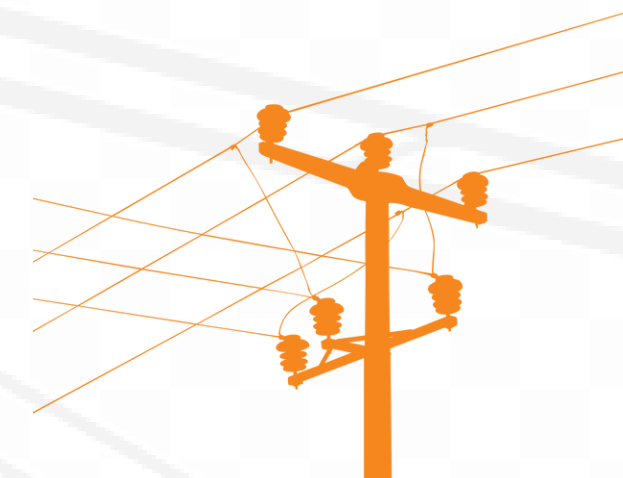
SOLAR POWER PLANT



INVERTER



ARISHIP ENERGY STORAGE SYSTEM



GRID

Analysis of the feasibility of the idea

THIS SYSTEM IS ESPECIALLY DESIGNED BY TAKING THE FEASIBILITY AND VIABILITY ON MIND BECAUSE WE HAVE VARIOUS OTHER GRID SCALE ENERGY STORAGE SYSTEM BUT THE MAJOR PROBLEM WITH THEM IS THEIR COST AND THEIR EFFICIENCY SOME OF THEM HAVE LOW COSTLY BUT THE EFFICIENCY IS ALSO LOW AND SOME OF THEM IS VERY EFFICIENT BUT THEY ARE COSTLY AND MORE COMPLEX OUR IDEA IS ALSO COSTLY BUT IT IS SIMPLE AND SUSTAINABLE FOR LONG PERIOD OF TIME WHICH IS GIVE US THE ROI IN LONG RUN

Potential challenges and risks

THE MAJOR CHALLENGES IN IMPLEMENTATION OF THIS SYSTEM IS CONTROLLING THE AIRSHIP AND INITIAL COST OF THE PROJECT AND THERE ARE NOT ANY MAJOR RISK WITH THIS SYSTEM

Strategies for overcoming these challenges

TO OVERCOME THE PROBLEM OF CONTROLLING OF THE AIRSHIPS WE NEED DIFFERENT TYPE OF GENERATOR WITH VERY COMPLEX AND STRONG GEAR SYSTEM AND WE ALSO NEED VERY STRONG INFRASTRUCTURE AND WE ALSO NEED TO DEVELOP SOME SPECIAL TYPE OF SECURITY SYSTEM THAT CAN HELP IN WORST CASE SCENARIOS AND WE ALSO NEED SKILLFUL WORKER TO MAINTAIN THIS TYPE OF SYSTEM

IMPACT AND BENEFITS

Potential impact on the target audience

THE IMPACT OF SYSTEM IS VERY HUGE ON THE PRESENT DAY ENERGY SYSTEM BEACUSE AT PRESENT TIME WE MAJORLY FACING TWO PROBLEMS THE POWER CONTRAILMENT AND SECOND THE GRID STABILITY THIS AIRSHIP ENERGY STORAGE SYSTEM SOLVE BOTH THE PROBLEMS

Benefits of the solution (social, economic, environmental, etc.)

THE MAJOR BENEFIT OF THE AIRSHIP ENERGY STORAGE SYSTEM IS THAT WE STORE THE WASTED ELECTRICITY AND THAT WITH ALSO REDUCE THE COST OF THE ELECTRICITY

SOME OTHER BANEFITS:

- 1. PREVENT SHORT TERM ENERGY CUT**
- 2. STORE THE ELECTRICITY FOR LONG PERIOD OF TIME**
- 3. MAINTAIN THE GRID STABILITY DUE TO ROTATING MASS**

OUR RESEACH PAPERS ON AIRSHIP ENERGY STORAGE SYSTEM

https://drive.google.com/drive/folders/1OlmwkSVISfpN04tMgmILq6QoycUJsi_9?usp=sharing

THANK YOU